Discrete Choice and Causal Inference

Consumer Choice, Program Evaluation, Policy Effects, and Regression Discontinuity

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Applied-Econometrics-with-Stata

Methods: alternative-specific conditional logit, propensity-score matching, difference-in-differences, event studies, and regression discontinuity.

Executive Summary

This portfolio project applies four econometric frameworks to distinct policy and consumer-choice questions. First, an alternative-specific conditional logit model examines household heating-system choices. Second, propensity-score matching evaluates the National Supported Work (NSW) job-training program using a nonexperimental CPS comparison group. Third, difference-in-differences and event-study specifications assess labor-market outcomes after implementation of the Americans with Disabilities Act (ADA). Fourth, the age-65 Medicare eligibility threshold is used in a regression-discontinuity design to study insurance coverage and health-care utilization.

The analysis emphasizes not only point estimates but also identification assumptions, specification limitations, and the distinction between descriptive associations and credible causal evidence.

Application	Primary method	Main finding
Heating-system choice	Alternative-specific conditional logit	Operating costs reduce choice probability more strongly than installation costs.
NSW job training	Propensity-score matching	Estimated ATET is about $494, but it is imprecise and statistically insignificant.
ADA labor-market effects	Difference-in-differences and event study	Post-ADA outcomes decline for disabled workers, especially weeks worked.
Medicare at age 65	Regression discontinuity	Insurance coverage jumps sharply at 65, with modest increases in doctor visits and hospital admissions.

1. Household Heating-System Choice

The first application models household choice among five heating-system alternatives. Utility depends on installation cost, annual operating cost, alternative-specific constants, and an idiosyncratic Type-I extreme-value error. The alternative-specific conditional logit framework uses 900 household choice occasions and 4,500 alternative-level observations.

1.1 Cost Sensitivity

Variable	Coefficient	Interpretation
Installation cost	-0.00153	Higher upfront cost reduces the probability that a system is selected.
Operating cost	-0.00699	Higher recurring cost substantially reduces choice probability.

Both cost coefficients are negative and statistically significant. The operating-cost coefficient is about 4.57 times as large in absolute value as the installation-cost coefficient. Interpreted as a tradeoff, households are willing to pay approximately $4.6 more in installation cost to reduce annual operating cost by $1, subject to the units used in the data.

1.2 Predicted Choice Shares

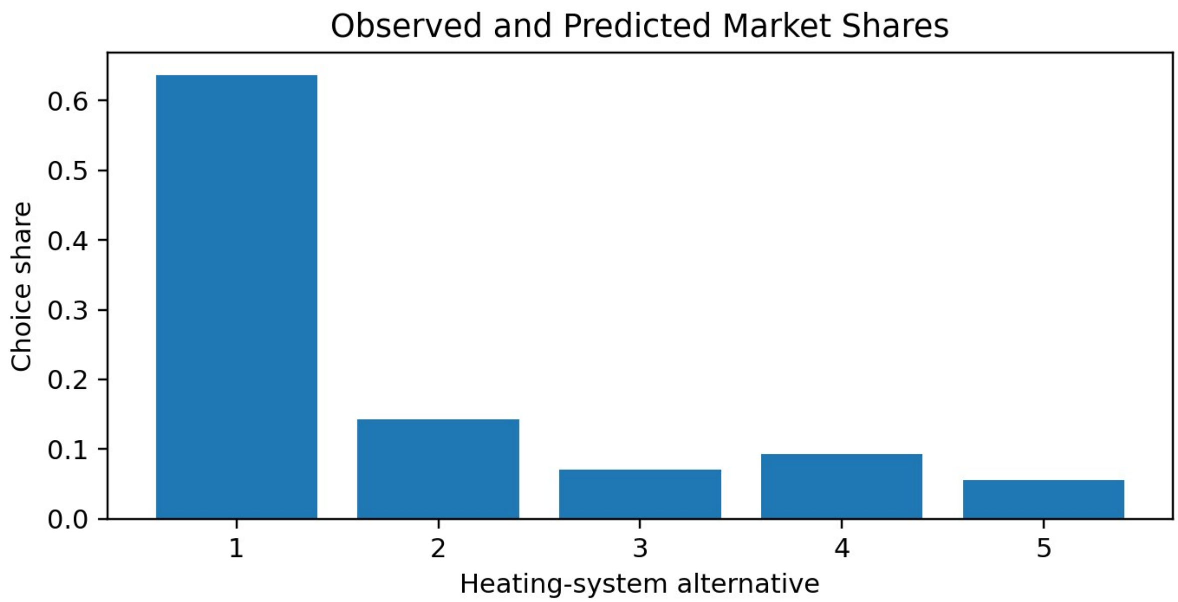

Figure 1. Predicted and observed market shares coincide because alternative-specific constants reproduce average choice frequencies.

Alternative	Predicted share	Observed share
1	0.6367	0.6367
2	0.1433	0.1433
3	0.0711	0.0711
4	0.0933	0.0933
5	0.0556	0.0556

Own partial effects are negative for every alternative. For alternative 1, a one-unit increase in installation cost lowers choice probability by about 0.035 percentage points, while a one-unit increase in operating cost lowers it by about 0.160 percentage points.

1.3 Income Heterogeneity

A richer specification interacts installation and operating costs with household income. Operating cost remains negative and significant (-0.00741), but installation cost becomes statistically insignificant. The interaction terms are not significant, so the sample provides limited evidence that price sensitivity varies systematically with income.

2. Job Training and Nonexperimental Comparison Groups

The NSW experimental treatment and control groups are broadly balanced, consistent with random assignment. In contrast, the CPS comparison sample differs sharply: CPS participants are older, more educated, less likely to be Black, more likely to be married, and have much higher pre-program earnings.

Characteristic	NSW treatment	NSW control	CPS
Average age	24.63	24.45	33.23
Years of education	10.38	10.19	12.03
Share Black	80.1%	80.0%	7.4%
Share married	16.8%	15.8%	71.2%

2.1 Identification Concern

When NSW treated participants are compared with CPS controls, unconfoundedness is doubtful. Even after conditioning on age, education, race, marital status, children, and prior earnings, unobserved differences in motivation, health, criminal history, family stability, and labor-market attachment may remain. The likely bias in a simple cross-sectional comparison is downward because NSW participants would probably have earned less than CPS workers even without treatment.

2.2 Propensity-Score Matching

Estimate	Value
ATET on 1978 earnings	$493.93
p-value	0.612
Statistical conclusion	Not statistically significant

The positive estimate is economically plausible, but the confidence interval is wide and includes zero. The result illustrates that matching on observed characteristics cannot fully overcome limited common support or unobserved differences between a disadvantaged program population and a general-population comparison group.

2.3 Difference-in-Differences as an Alternative

Using 1974 earnings as a pre-treatment outcome and 1978 earnings as a post-treatment outcome can remove time-invariant group differences. However, identification then depends on parallel earnings trends between NSW participants and CPS workers in the absence of treatment. Given their markedly different labor-market trajectories, that assumption remains demanding.

3. Labor-Market Effects of the Americans with Disabilities Act

The ADA analysis treats disabled workers as the treatment group, nondisabled workers as the comparison group, and 1992 onward as the post-policy period. The coefficient on the disabled-by-post interaction is interpreted as the difference-in-differences effect.

Outcome	Baseline DiD	Adjusted DiD	Inference
Weeks worked	-26.84	-26.53	Statistically significant at the 1% level
Weekly earnings	-$130.05	-$119.03	Statistically significant at the 1% level

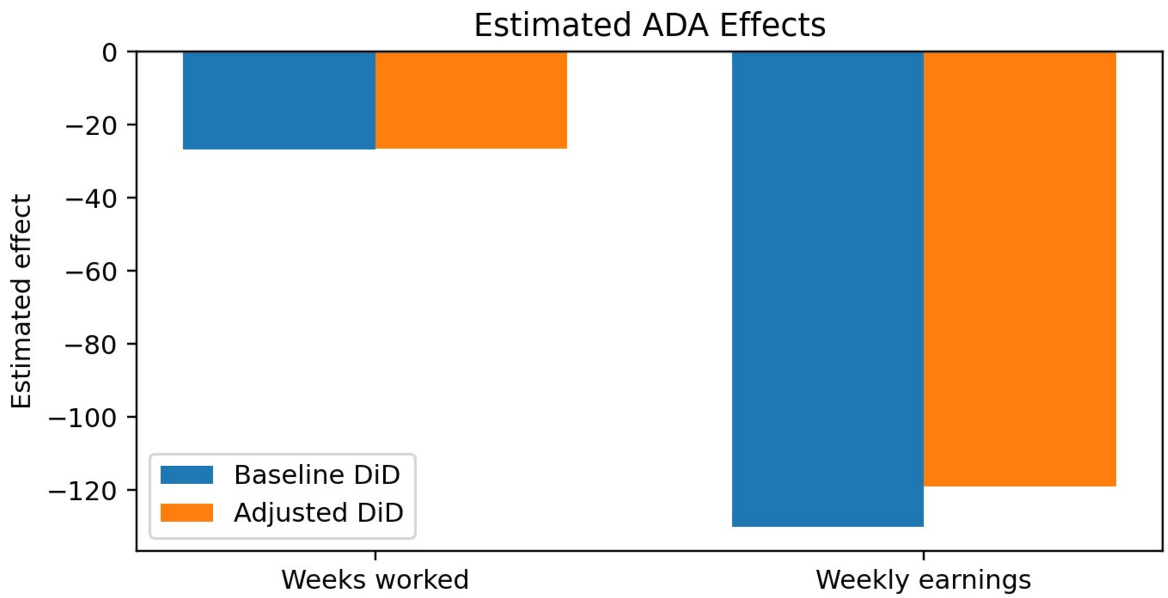

Figure 2. Baseline and covariate-adjusted ADA estimates are similar in magnitude.

The estimates suggest that labor-market outcomes for disabled workers declined after ADA implementation relative to nondisabled workers. One possible mechanism is that employers responded to expected accommodation, compliance, or dismissal costs by reducing hiring, work opportunities, or compensation. This interpretation should remain cautious because heterogeneous trends and omitted policy changes may also affect the estimates.

3.1 Event-Study Evidence

In the dynamic specification, 1991 is the omitted baseline year. Pre-policy coefficients for weeks worked are close to zero, providing no strong visual evidence against parallel trends. Post-policy employment effects become progressively more negative, suggesting that adjustment occurred gradually. Weekly-earnings coefficients are more volatile and less precise, indicating a clearer effect on employment intensity than on wages.

4. Medicare Eligibility at Age 65

Age 65 creates a sharp eligibility threshold for Medicare. This institutional rule provides a quasi-experimental source of variation in insurance coverage and motivates a regression-discontinuity design.

Coverage type	Below age 65	Age 65 and older	Change
Medicare	7.7%	87.3%	+79.6 pp
Private insurance	74.9%	62.4%	-12.5 pp
Medicaid	5.7%	8.9%	+3.2 pp
Any insurance	84.0%	97.5%	+13.5 pp

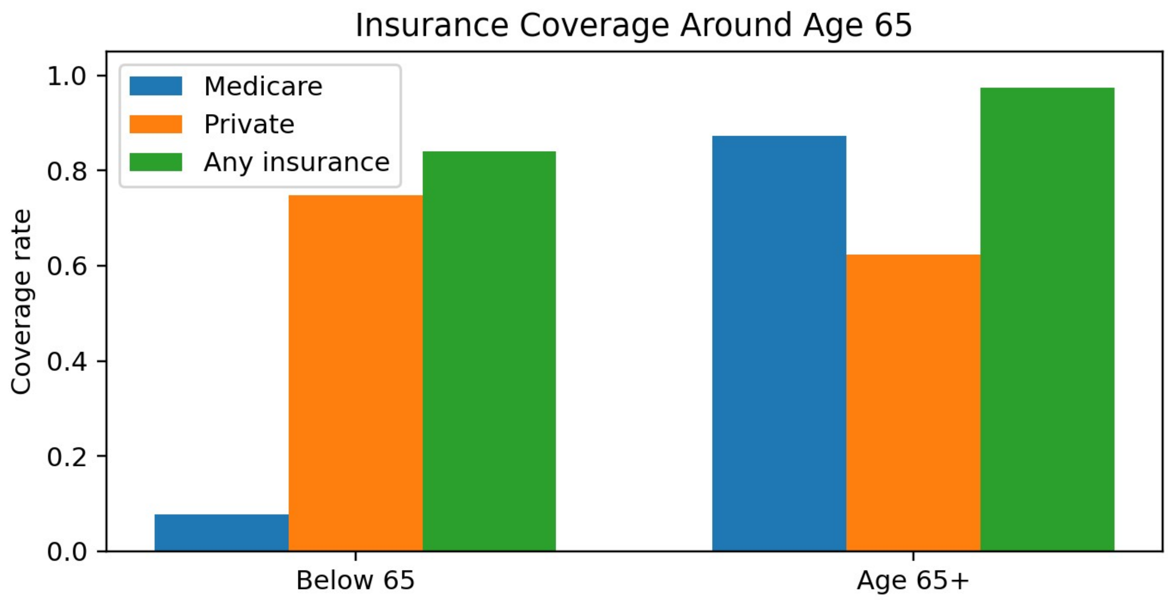

Figure 3. Medicare eligibility sharply expands public and total insurance coverage, while private coverage declines.

4.1 OLS Associations

Outcome	Insurance coefficient	Interpretation
Doctor visit	0.192	Insured individuals are 19.2 percentage points more likely to report a doctor visit.
Hospital admission	0.044	Insured individuals are 4.4 percentage points more likely to report hospitalization.

These OLS estimates should be interpreted as associations, not causal effects. Insurance status is endogenous: health needs, employment, income, risk preferences, and access to care jointly influence both coverage and utilization.

4.2 Reduced-Form Regression Discontinuity

Outcome	Age-65 discontinuity	p-value
Doctor visit	+0.0213	0.037
Hospital admission	+0.0142	0.039

Turning 65 increases the probability of a doctor visit by about 2.1 percentage points and hospital admission by about 1.4 percentage points. These reduced-form effects are consistent with Medicare eligibility increasing health-care utilization through expanded insurance coverage. A full fuzzy-RDD or instrumental-variable analysis would divide these outcome discontinuities by the first-stage jump in insurance coverage.

5. Methodological Lessons and Limitations

- Conditional-logit coefficients identify utility tradeoffs under the independence-of-irrelevant-alternatives assumption; alternative substitution patterns may be too restrictive.
- Propensity-score matching balances observed covariates but cannot remove bias from unobserved differences or weak common support.
- Difference-in-differences requires parallel counterfactual trends. Event studies are useful diagnostics, but near-zero pre-trends do not prove the assumption.
- Regression discontinuity identifies a local effect near the cutoff and requires smooth potential outcomes and no precise manipulation of the running variable.
- Binary-outcome OLS estimates require robust standard errors and should be interpreted carefully when used as linear probability models.

6. Reproducibility

The accompanying Stata do-file reproduces the analysis with relative data paths, input validation, automated package checks, stored model estimates, exported figures, and saved derived data. The original datasets are not included because they may be subject to redistribution restrictions.

Recommended repository structure:

```
04_Discrete_Choice_and_Causal_Inference/
  README.md
  04_discrete_choice_causal_inference.do
  Discrete_Choice_and_Causal_Inference_Report.pdf
  data/README.md
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Conclusion

Across four applications, the project demonstrates how econometric design shapes substantive conclusions. Recurring operating costs dominate household heating-system choices; matching a disadvantaged training population to general-population controls remains difficult; ADA estimates indicate post-policy declines in disabled workers' labor-market outcomes, especially employment intensity; and Medicare eligibility at age 65 produces a sharp coverage expansion accompanied by modest increases in health-care use. The analysis shows the value of combining model estimates with explicit discussion of identification, diagnostics, and limitations.